

Large Language Model Persona Vector Extraction Technology and Its Application Concepts in Psychology Cross-Research

Technical Disclosure and Defensive Publication

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Abstract

Recent advances in the interpretability of large language models (LLMs), particularly Anthropic's feature steering and persona vector extraction techniques, have provided powerful

tools for understanding high-level semantic representations within models. The core principle is that high-level semantic concepts—such as personality traits, emotional tendencies, and values—manifest as linear directions in the high-dimensional activation spaces of neural networks. This means that abstract psychological concepts can be mathematically modeled and computationally operated upon like geometric vectors.

Based on this principle, this paper proposes two novel methods for AI-psychology cross-research: (1) **Systematic investigation of the mathematical properties of persona vectors**—quantitative analysis of "personality traits" as a core psychological construct, exploring the geometric relationships between opposing personality pairs and orthogonal personality pairs, and examining the internal consistency and rigor of mainstream personality taxonomies (such as the Five-Factor Model and the 16PF) within the model; and (2) **Mapping technology for projecting real human personalities into LLM representation spaces**—transforming an individual's psychological traits into a "personality digital twin vector set" within the model's vector space. This paper elaborates on three technical sampling frameworks (dimension-based sampling via system prompts, data-driven sampling via user inputs, and structured task sampling), and uses the 16PF as an illustrative example to demonstrate the feasibility of digitizing real human personality—the vectorizability verification for this example has been completed experimentally. Based on these foundations, an iterative path under the finite element approach and a "three-step" implementation plan are proposed. The overall technical framework is in a state ready for practical implementation.

Keywords: Persona Vector; Feature Steering; Orthogonal Persona; Personality Digitization; Digital Twin; Sampling Protocol; Computational Psychology

1. Background and Technical Foundations

Recent advances in the interpretability of large language models have revealed the correspondence between internal neural activation patterns and high-level semantic concepts. Anthropic's research team has made significant breakthroughs in this direction, successively proposing Feature Steering [1] and Persona Vector [2] extraction techniques. The core principle is that high-level semantic concepts—such as personality traits, emotional tendencies, and values—manifest as linear directions in the high-dimensional activation spaces of neural networks. This means that abstract psychological concepts can be mathematically modeled and computationally operated upon like geometric vectors.

This discovery opens a new path for cross-research between psychology and artificial intelligence: if personality traits can be represented and manipulated as vectors within models, then "digital twins of personality" become technically feasible.

2. Two Cross-Research Methods

Based on the above principles, two novel methods for AI-psychology cross-research are proposed:

2.1 Method 1: Systematic Investigation of the Mathematical Properties of Persona Vectors

The purpose of this method is to conduct quantitative analysis of "personality traits" as a core psychological construct. Specific contents include: exploring the geometric relationships between opposing personality pairs (e.g., extraversion-introversion) and orthogonal personality pairs (e.g., independence-agreeableness) in vector space, and examining the internal consistency and rigor of mainstream personality taxonomies (such as the Five-Factor Model and the 16PF) within the model.

Preliminary technical validation and mathematical derivation for this method have been completed, with results highly consistent with theoretical expectations. This indicates that: (1) the technical conditions for "human personality digital twin technology" are relatively mature on the LLM side; and (2) several classical constructs in psychology may acquire new, computable empirical analytical frameworks through this approach.

2.2 Method 2: Mapping Technology for Projecting Real Human Personalities into LLM Representation Spaces

This method aims to map an individual's psychological traits into the vector space of an LLM, thereby constructing that individual's "personality digital twin vector set." Once obtained, this vector set enables rigorous mathematical analysis of the individual's personality characteristics, such as comparing vector distances and clustering relationships between different individuals, predicting their response patterns in specific contexts or tasks, and even exploring the dynamic patterns of personality states as input conditions vary.

3. Sampling Framework Technical Concepts

Regarding specific sampling and mapping protocols, three concepts are currently proposed:

3.1 Scheme 1: Dimension-Based Sampling via System Prompts

The core logic of Anthropic's original technique lies in feeding both the "system prompt" and the "user input" into the model jointly, and preserving the net effect of the system prompt on the overall vector through differential computation. This means that by describing a particular personality trait in detail within the system prompt, the corresponding persona vector can be extracted.

Accordingly, one can select a set of personality-descriptive categories with orthogonal mathematical properties (such as the dimensions of the Five-Factor Model), conduct qualitative sampling judgments on the subject for each dimension, form a textual description of their personality and place it in the system prompt, then perform differential comparison against an empty or neutral system prompt. The resulting output from running the procedure is the individual's "persona vector."

This scheme is relatively close to implementation, with intuitive sampling design. As a verification instance for the low-dimensional sampling approach, the vectorizability of the 16PF has been experimentally validated, with results highly consistent with expectations.

3.2 Scheme 2: Data-Driven Sampling via User Inputs

Personality information is organically embedded in the "user input" portion (rather than being copied into the system prompt or mechanically transferring system prompts to user inputs), and a sufficiently rich sample set that effectively distinguishes the individual's personality characteristics is collected through extensive sampling. This is input into the LLM and differentially compared against a set of neutral control samples, thereby extracting the individual's "persona vector."

The advantage of this scheme is its ability to break through preset dimensional limitations, allowing personality traits to emerge naturally from the data. The drawback is the requirement for substantial data collection and the design of an effective neutral control group.

3.3 Scheme 3: Structured Task Sampling (Compromise Approach)

If Scheme 2 is temporarily infeasible, a structured sampling protocol can be designed, such as asking subjects to describe a specific painting within a word limit, retell a specific passage, evaluate a specific viewpoint, or describe/evaluate a specific character. These responses are then differentially compared against a neutral control group (e.g., an objective description of the painting generated by an AI) to extract the corresponding vector set.

It should be noted that whether this vector set strictly equates to Anthropic's defined Persona Vector requires further 论证, but it at least provides an operable analytical object for quantitative research.

4. Feasibility Demonstration: A Validated Instance

To demonstrate the feasibility of the above technical concepts, the 16PF (16 Personality Factor Questionnaire) is selected as a validation instance for the low-dimensional sampling approach.

The validation objectives are to confirm: (1) whether the basic dimensions of the 16PF can be mapped to corresponding vector directions in the LLM representation space; (2) whether the dimensional vectors satisfy mathematical orthogonality; and (3) whether the persona vectors generated through linear combination of these orthogonal basis vectors can accurately reproduce the personality types defined by the 16PF classification.

All three levels of validation have been completed successfully, with experimental results highly consistent with theoretical expectations:

1. Each basic dimension yields stable corresponding vector directions in the target LLM's representation space;
2. Orthogonality among the basic dimensional vectors has been verified experimentally;
3. Linear combinations of the orthogonal basis vectors accurately reproduce the corresponding personality types.

This validated instance provides principle-level feasibility evidence for the "digitization of real human personality" technical pathway—at least for personality models that have been structurally described, precise mapping into LLM representation spaces is achievable. This provides an experimental foundation for subsequent research. However, the methods described in this paper are not limited to the 16PF framework; other mainstream personality taxonomies can undergo similar validation and application within this framework.

5. Iterative Path Under the Finite Element Approach

Drawing on the finite element analysis methodology, this paper proposes an iterative optimization path from low-dimensional to high-dimensional:

First-tier scheme compresses sampling dimensions to the extreme (e.g., using the already-validated 16PF questionnaire as an instance), offering high predictability of subjects and strong practical feasibility.

Subsequent iterations follow the finite element principle, progressively identifying and adding other sampling dimensions that influence personality. As dimensionality N increases, the fidelity of the digital twin to the real person continues to improve.

The core advantage of the digital twin lies in its computability, not its initial fidelity. Fidelity is strongly correlated with dimensionality N ; high expectations should not be set in early stages. Only under higher-tier (data-driven) sampling schemes can one reasonably expect LLM outputs to be highly consistent with the real person.

Several existing sampling protocols with varying dimensionalities are currently available for feasibility demonstration purposes.

6. "Three-Step" Implementation Plan

Based on the above analysis, the following implementation plan is proposed:

Step 1 (One-time foundational work): Identify the N most impactful sampling dimensions and establish the basis vectors—requires specialized knowledge, but only needs to be done once. The completed 16PF experiment has demonstrated the operational feasibility of this step.

Step 2 (Moderate difficulty): Design a survey questionnaire to quantitatively sample each dimension—also requires professional design capability.

Step 3 (Extremely easy to operate): Use the collected data as coefficients to linearly combine and generate the person's digital twin.

7. Hybrid Augmentation Approach

On top of the first-tier scheme, a hybrid approach can be superimposed: **persona vector method + fine-tuning**, to enhance fidelity and compensate for precision limitations caused by insufficient initial dimensionality.

Specifically: based on the extracted persona vector, use a small amount of the subject's natural language data to perform lightweight fine-tuning of the model, bringing the model's output style and content preferences closer to the subject themselves. This approach maintains the interpretability of the persona vector framework while leveraging fine-tuning to capture higher-dimensional personalized features.

8. Discussion

8.1 Technical Feasibility Summary

The two methods and their supporting sampling schemes proposed in this paper are technically feasible:

- **Method 1 (systematic investigation of persona vector mathematical properties)** has largely completed its theoretical derivation; the construction method for orthogonal personas is mathematically self-consistent;
- **Method 2 (mapping real human personalities)** offers three categories of sampling schemes covering different needs from rapid deployment to high precision, with feasibility supported by experimental instances.

8.2 Status of Completed Validation

In the validation experiment using the 16PF as an instance, vectorizability of the basic dimensions has been confirmed experimentally, orthogonality among dimensions has been verified, and combinatorial reproduction is consistent with theoretical expectations. This validation provides experimental-level feasibility evidence for the technical pathway of Scheme 1, but the applicability of the methods described herein is not limited to this particular framework.

8.3 Key Outstanding Issues

The following core challenges remain for subsequent research:

1. Designing sampling protocols with sufficient psychological validity;
2. Performing interpretable semantic annotation of extracted vector sets, ensuring verifiable correspondence with their theoretical constructs;
3. Establishing a complete mathematical 论证 framework to ensure reproducibility and reliability;
4. The issue of unified multimodal input representation—current methods are primarily text-based;
5. Further empirical validation of Scheme 2 (data-driven sampling) and Scheme 3 (structured task sampling).

8.4 Future Research Directions

The following directions are recommended for subsequent research:

1. Extending the validated methodological framework to other mainstream personality taxonomies such as the Five-Factor Model;

2. Collecting existing sampling protocols with varying dimensionalities and conducting systematic comparative experiments;
3. Exploring cross-validation between Method 1 and Method 2—comparing the orthogonal personality structures extracted from AI models with clustering patterns observed in human data;
4. Advancing the experimental design for Scheme 2 (data-driven sampling) to explore higher-dimensional pathways for personality digitization.

9. Conclusion

This paper has proposed two novel methods for AI-psychology cross-research based on Anthropic's feature steering and persona vector extraction techniques, and has elaborated on sampling scheme concepts.

Method 1 (systematic investigation of the mathematical properties of persona vectors) leverages linear algebraic tools for quantitative analysis of personality traits, providing an experimental framework for exploring the independent dimensions of personality structure through orthogonal persona construction. **Method 2 (mapping real human personalities into LLM representation spaces)** transforms an individual's psychological traits into computable vector forms, enabling digital archiving and quantitative comparison of personality.

Regarding sampling schemes, three technical concepts are proposed—dimension-based sampling via system prompts (Scheme 1), data-driven sampling via user inputs (Scheme 2), and structured task sampling (Scheme 3)—covering different pathways from rapid deployment to high precision.

Using the 16PF as a validation instance for the low-dimensional sampling approach, its vectorizability has been experimentally validated. This instance provides experimental-level feasibility evidence for the technical pathway described herein. However, the methods proposed in this paper constitute a general framework and are not limited to any specific personality classification system.

Overall, the methods described in this paper are theoretically self-consistent, technically feasible, with core validation work preliminarily completed, and the overall technical framework is in a state ready for practical implementation, meriting follow-up empirical research.

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